INCH-POUND

MIL-C-50652/14 (AR)
AMENDMENT 4
8 February 1993
SUPERSEDING
AMENDMENT 3
20 May 1976

MILITARY SPECIFICATION SHEET

CIRCUIT CARD ASSEMBLY: 10548359

(CONVERTER, RANGE D/A, 4A16)

This amendment forms a part of MIL-C-50652/14 (MU), dated 15 April 1971, and is approved for use by the US Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

Table I, Input voltages, add:

"-10.0 ± 0.1 VDC	P1-12
-10.0 V Return	P1-13"

3.5.4: Delete in its entirety and substitute:

"3.5.4 **DC** Offset - With the logic levels applied as specified in Step **12** of Table II, the output voltage at P1-41 shall be. 0.000 ± 0.002 Vdc at ambient temperature.

3.5.4.1 Offset Thermal Drift - The DC OFFSET shall exhibit a thermal drift of ± 0.002 Vdc maximum over the operating temperature range specified by MIL-C-50652.

3.5.4.2 Offset Adjust - The output voltage at P1-41 shall be adjusted to 0.000 ± 0.050 mvdc by adjusting an external 5000 ohm Potentiometer that is probe connected to the BAL/ARM/BAL terminals of AR2."

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3.5.5: Delete in its entirety and substitute:

"3.5.5 Gain Adjust - With the logic levels applied as specified in Step 1 of Table II, adjust R4 to obtain a ratio of the voltage at P1-41 to the reference voltage at P1-39 of **1.00000 ± 0.00003.**

3.5.5.1 Gain Thermal Drift - The gain shall exhibit a thermal drift ratio of ±. 0.001 maximum over the operating temperature range specified by **MIL-C-50652."**

3.5.6: Delete in its entirety and substitute:

"3.5.6 Linearity - With the logic levels applied as specified in Steps 2 through 11 of Table II, the ratio of the voltage at P1-41 to the reference voltage at P1-39 shall be as specified by Table II ± 0.00025. Adjust OFFSET and GAIN prior to measurement."

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PAGES 4, 5 and 6

Tables II, III and IV: Delete and substitute:

TABLE II - LINEARITY RATIOS

STEP	RANGE (METER)	INPUT LOGIC LEVEL AT TERM TERMINALS P1- 9	11	18	17	2	4	15	5	6	33	RATIO
1	5000.	0	0	0	1	0	1	1	1	1	1	
2	2560	0	0	0	0	0	0	0	0	0	1	0.51200
3	1280	0	0	0	0	0	0	0	0	1	0	0.25600
4	640	0	0	0	0	0	0	0	1	0	0	0.12800
5	320	0	0	0	0	0	0	1	0	0	0	0.06400
6	160	0	0	0	0	0	1	0	0	0	0	0.03200
7	80	0	0	0	0	1	0	0	0	0	0	0.01600
8	40	0	0	0	1	0	0	0	0	0	0	0.00800
9	20	0	0	1	0	0	0	0	0	0	0	0.00400
10	10	0	1	0	0	0	0	0	0	0	0	0.00200
11	5	1	0	0	G	0	0	0	G	0	0	0.00100
12	0	0	0	0	0	0	0	0	0	0	0	

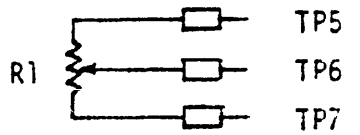
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Table V, Item number 6: Delete in its entirety and substitute:

Visual Display:	5 1/2 Digits
Input Impedance:	100 Kilohm Min
DC Ratio:	± 1: 1
Accuracy:	± 0.007% RDG ± C.001% FS
Ratio Reference Voltage:	10.2 Vdc Min
CMRR:	120 dB minimum for DC 120 dB minimum for 60 Hz

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Figure 1, ADD:



Change J1-41, J1-39 connection:

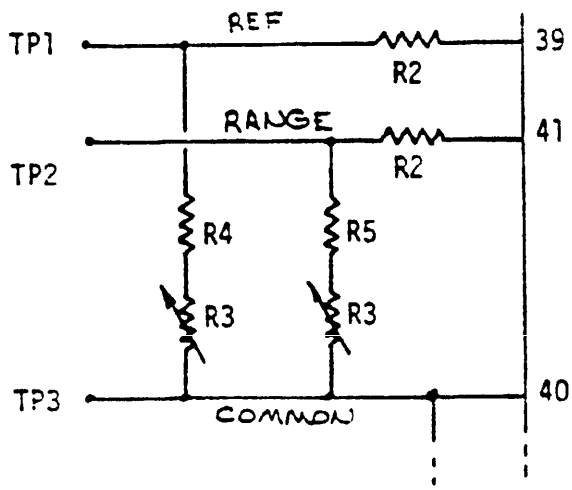


Table VI: Add the following:

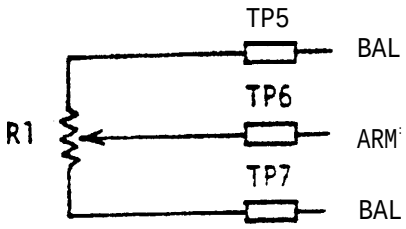
R1	5KΩ , ±5% 5 - Turn Pot
R2	10.0KΩ, ± 1% Resistor
R3	500 ohm, ± 3% 10 - Turn Pot
R4, R5	Nominal 10KΩ, select to obtain ± 0.01% Resistive divide by two networks when Item 6 is connected to TP1, TP2 and TP3 (adjust R3 as required)
TP5, TP6, TP7	Spring Loaded PCB Probe

4.6.1: Change MIL-C-59652 to MIL-C-50652.

4.6.3: Line 4, delete: "through IV".

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Figure 2: **Add** the following:



AR2 terminals, UUT

4.6.3.4: Delete in its entirety and substitute:

"***4.6.3.4*** DC Offset - Using switches S1 through S10 apply logic levels specified in Step 12, Table II, and with R1 disconnected from AR2, measure the DC OFFSET voltage between **TP2** and **TP3. It shall be** as specified by 3.5.4.

4.6.3.4.1 Offset Adjust - Connect R1 to AR2 and adjust to obtain a DC OFFSET as specified by 3.5.4.2.

4.6.3.4.2 Offset Thermal Drift - The DC OFFSET shall exhibit a thermal drift specified by 3.5.4.1."

4.6.3.5: Delete in its entirety and substitute:

"4.6.3.5 Gain Adjust - Using switches S1 through S10 apply **logic levels** specified in Step 1, Table II, and with DC OFFSET adjusted as specified by 4.6.3.4.1, adjust R4 as specified-by 3.5.5.

4.6.3.5.1 Gain Thermal Drift - The gain shall exhibit a thermal drift as specified by ***3.5.5.1.***"

4.6.3.6: Delete in its entirety and substitute:

"4.6.3.6 Linearity - Using switches S1 through S10 **apply logic levels specified in 3.5.6. Use the Ratio** Voltmeter, Item **6, to** measure the ratio of **TF2 to TP1. The** ratio shall be as specified by 3.5.6."

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Due to the extensiveness of the changes, the margins of this amendment are not marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made.

Custodian:
Army-AR

Preparing activity:
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(Project 1285-A883)

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